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Unit-IV
Chemistry of nanomaterials

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Outline

Sl No	Topic to be taught
1	Introduction to Nano materials
2	Classification and Size dependent properties
3	Surface area, Catalytic properties, Optical properties
4	Synthesis of nano materials, solution combustion method
5	Sol-gel method
6	Synthesis, properties and applications of carbon nano tubes.
7	Synthesis, properties and applications of graphene
8	Nano lubricants: Types of nanoparticles as lubricant additives and their application in defense, automobile and spacecrafts.

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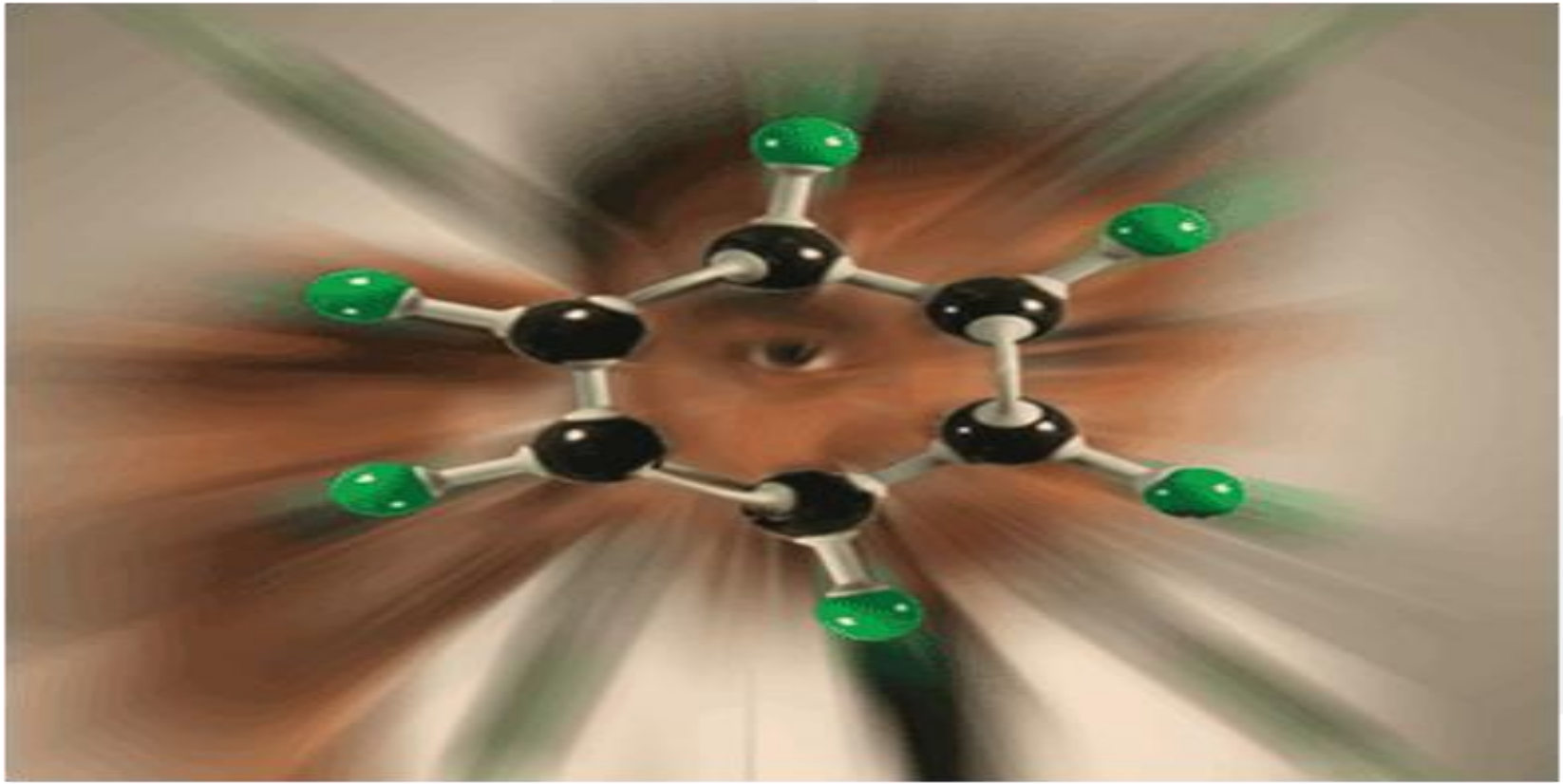
What is nano technology?

Chemistry?



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What is nano technology?



"Nanotechnology is the art and science of manipulating matter at the nanoscale"

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How small is nano scale?



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How small is nano scale?



A nanometer is one billionth of a meter



DNA Sample: Approx.**2 nm**

Human Hair: Approx. **$1x10^5$ nm**

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What are nano materials?

- **Nanomaterials** are defined as materials with an average grain size less than 100 nanometers.

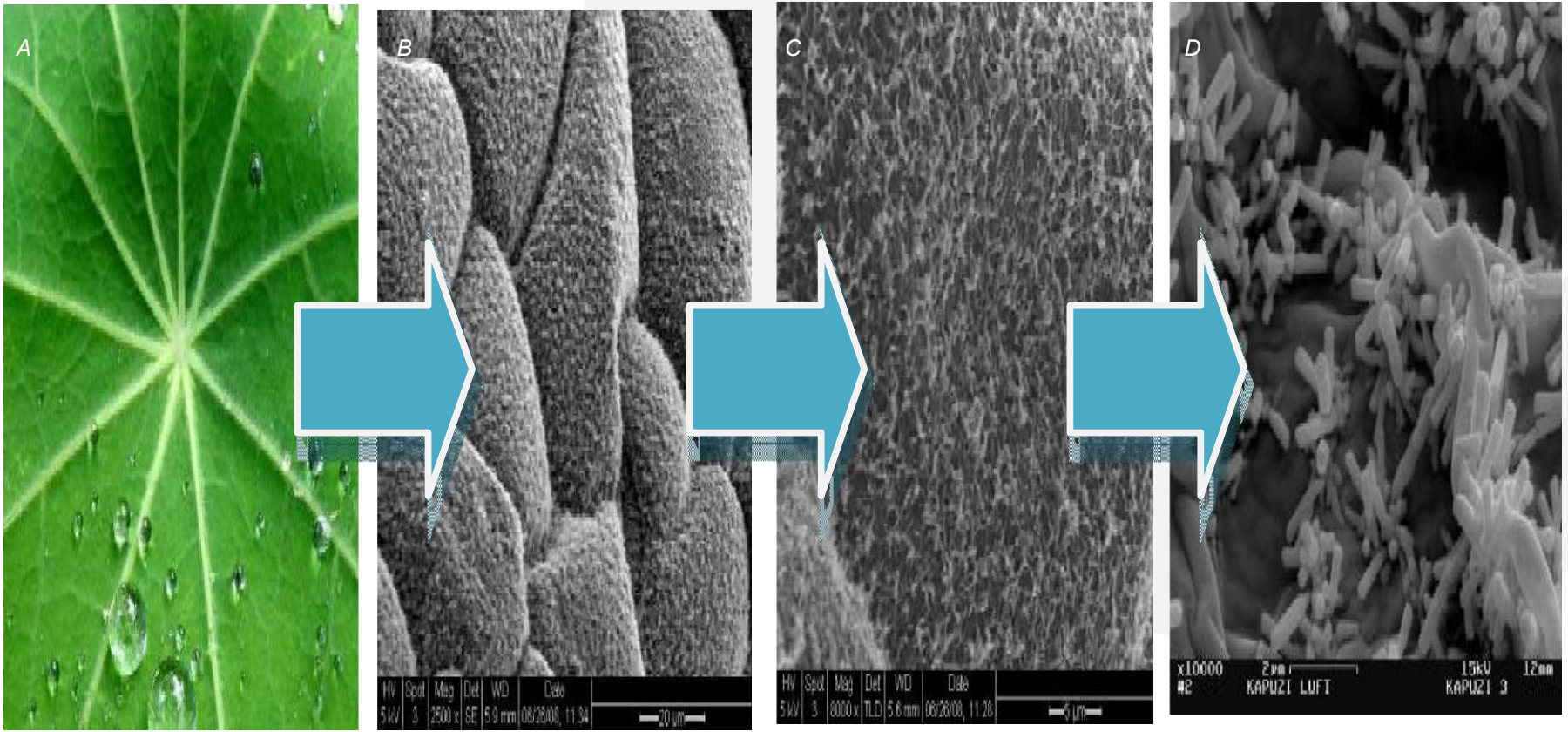
- In scientific language it is a prefix which has a value equal to "one billionth, i.e. 10^{-9} Therefore, one nanometer is one billionth of a meter (1 nm = 10^{-9} m).

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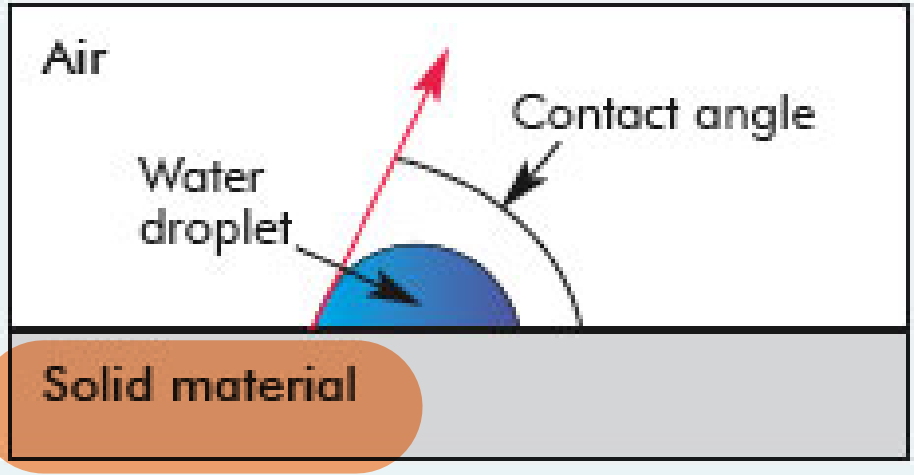
Learning from the nature?

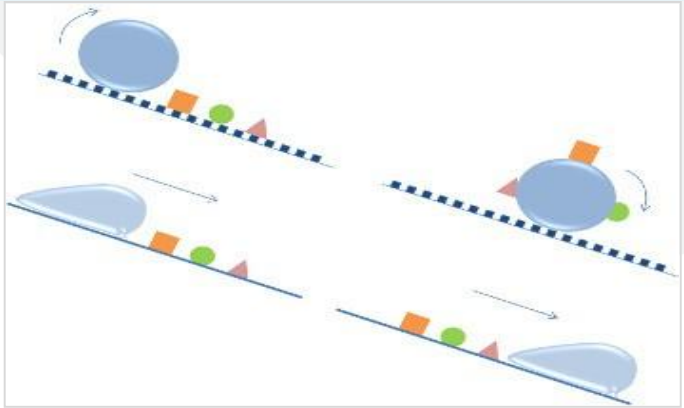
Lotus effect

- W. Barthlot….1997……..self cleaning properties….Water repellent property

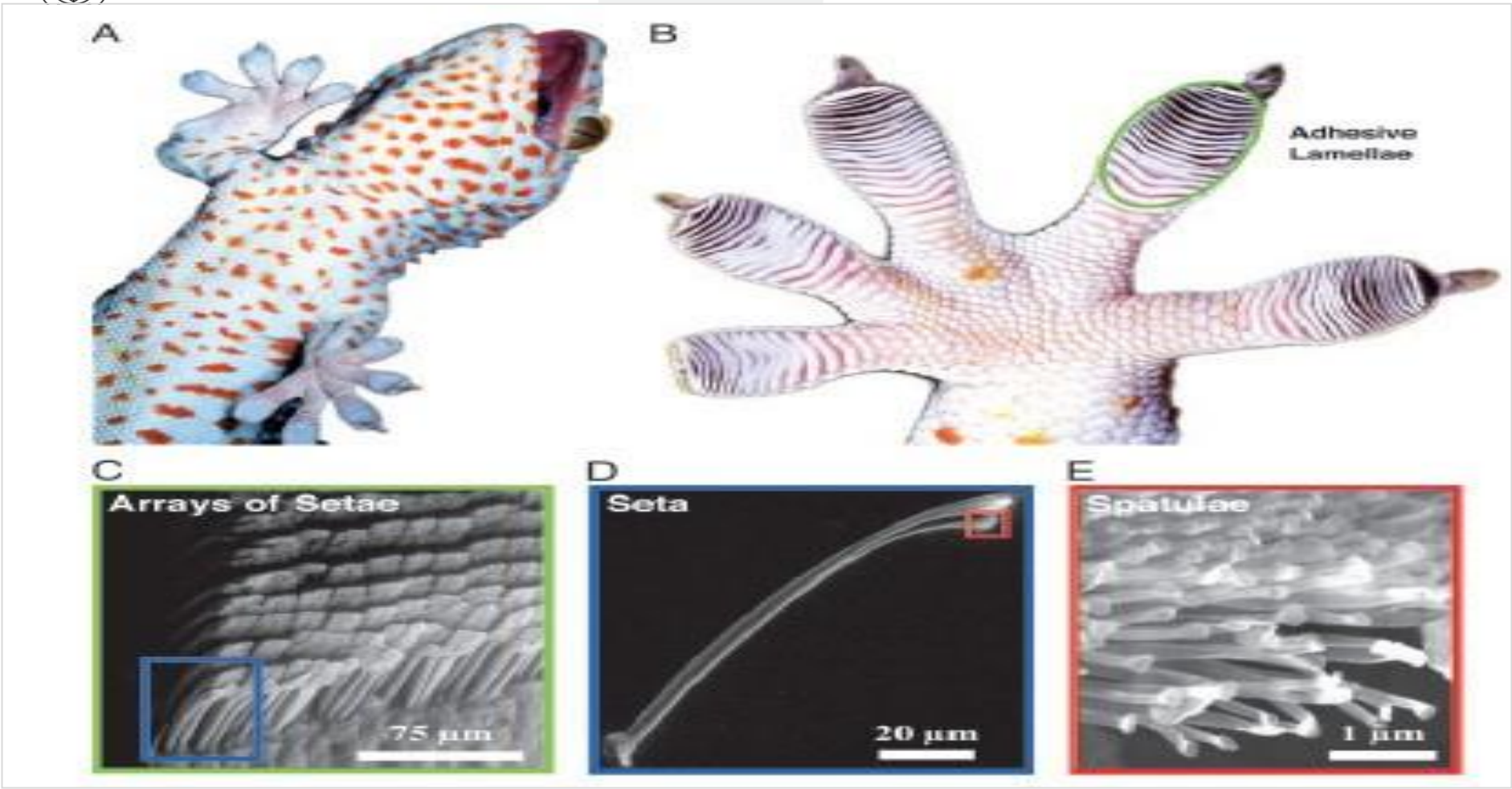





Air
Contact angle
Water droplet
Solid material



Gecko climbing

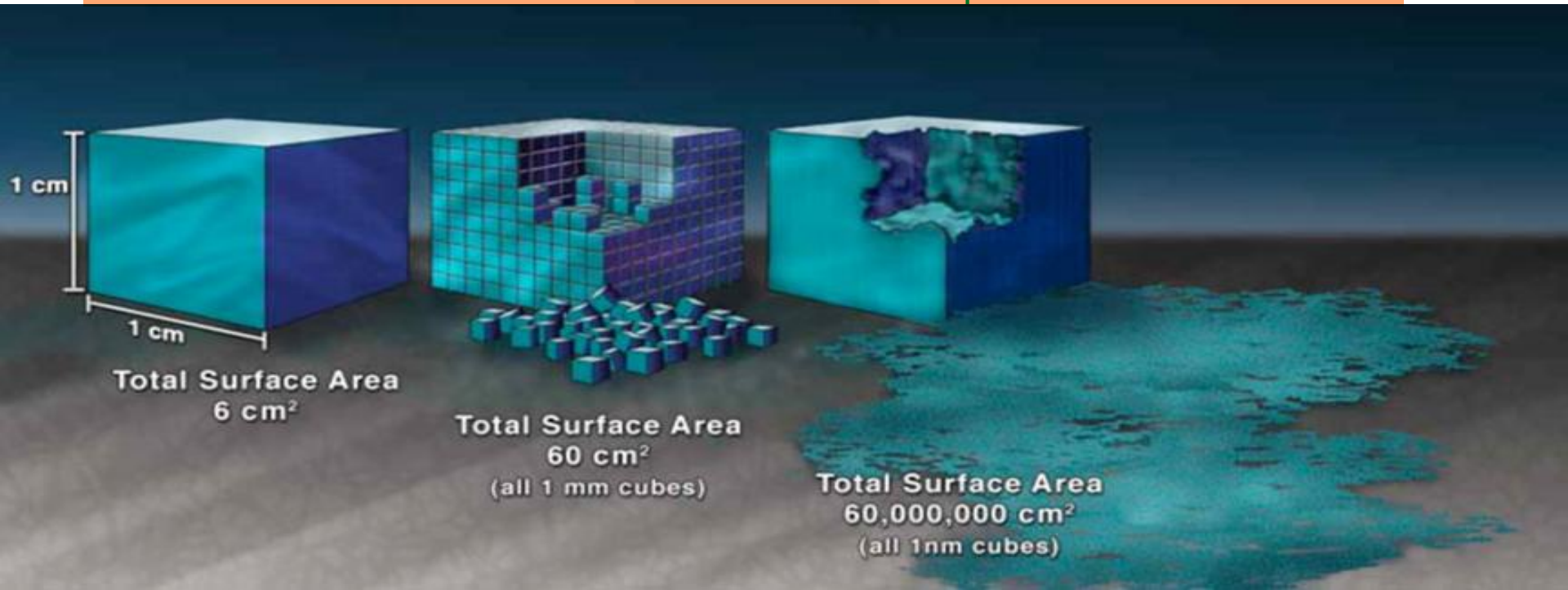


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Properties of nano materials

Surface area:

- As the size of particle decreases the surface area increases.
- Surface area to volume ratio increases as the particle size decreases.



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Properties of nano materials

Catalytic Properties:

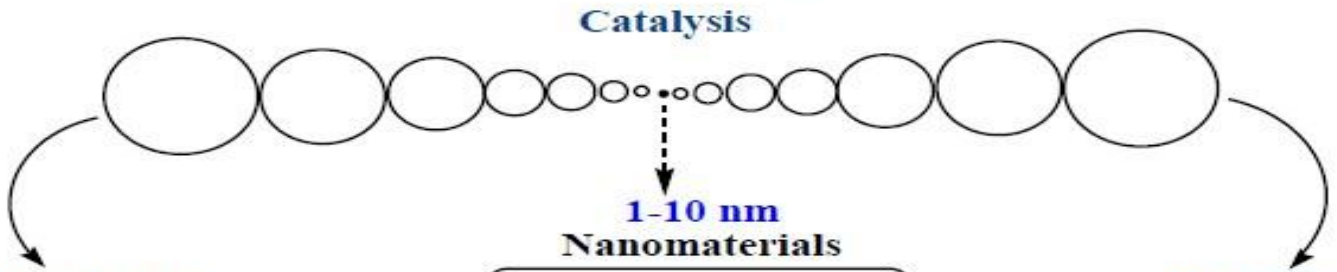
- Nanoscale catalysts have high specific surface area and surface energy, which ultimately lead to the high catalytic activity.
- Nano-catalyst improves the selectivity of the reactions by allowing reaction at a lower temperature, reducing occurrence of side reactions, higher recycling rates and recovery of energy consumption.
- Therefore, these are widely used in green chemistry, environmental remediation, efficient conversion of biomass, renewable energy development and other areas of interest.

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Properties of nano materials

Catalytic Properties:

Catalytic activity of materials-How it is depend on the size of materials?



Homogeneous Catalysis

Merits:

1. High activity
2. High chemo-and regio-selectivity

Demerits:

Cumbersom product purification and difficulty in catalyst recovery

Nano-Catalysis

- High activity
- High selectivity
- Excellent stability
- Easily separable
- Energy efficient
- Atom economy

Heterogeneous Catalysis

Merits:

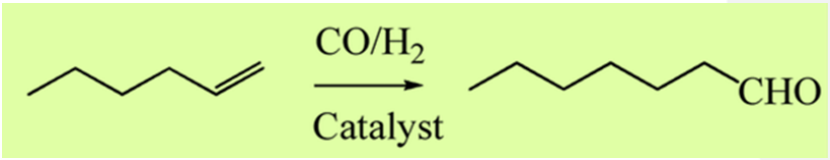
1. Excellent stability
2. Easy accessibility
3. Easily seperable

Demerits:

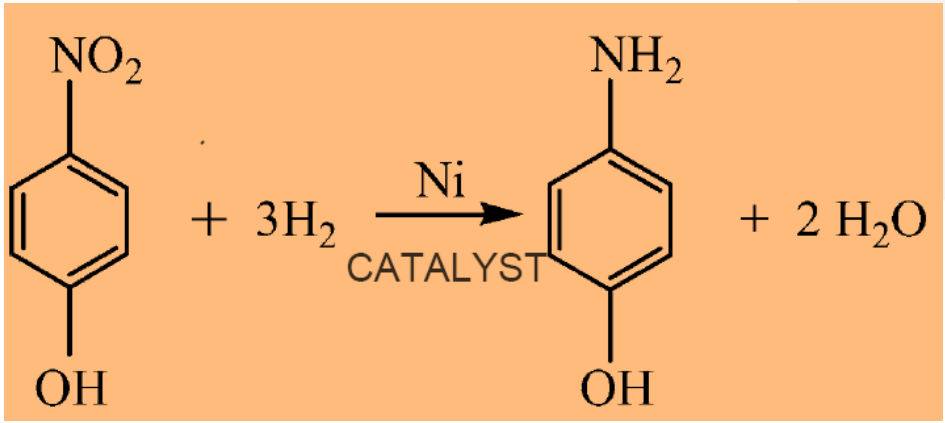
Inferior catalytic activity relative to their counterpart homogeneous and requires more reaction time.

Properties of nano materials

Catalytic Properties:



The highest activity with 100% total conversion and 96% yield of aldehydes was obtained with the Rh/nano- ZnO catalyst.



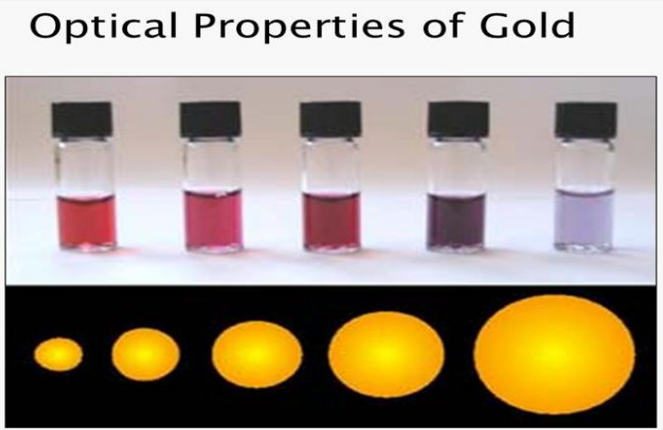
These results indicate that the catalytic stability of **nano-sized nickel** system is superior to that of the **Raney nickel system**

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Properties of nano materials

Optical property: Surface Plasmonic Resonance (SPR)

- Small molecules absorb light, leading to some color. It is a size dependent property.
- Gold which readily forms nano-particles but not easily oxidised, exhibits different colors depending on the size.
- Gold colloids have been used to make color glasses. Ruby glass contains finely dispersed gold colloids.
- Silver and copper also give attractive colors.



Properties of nano materials

Optical property: Surface Plasmonic Resonance (SPR)



The gold we know:



The gold we are discovering:

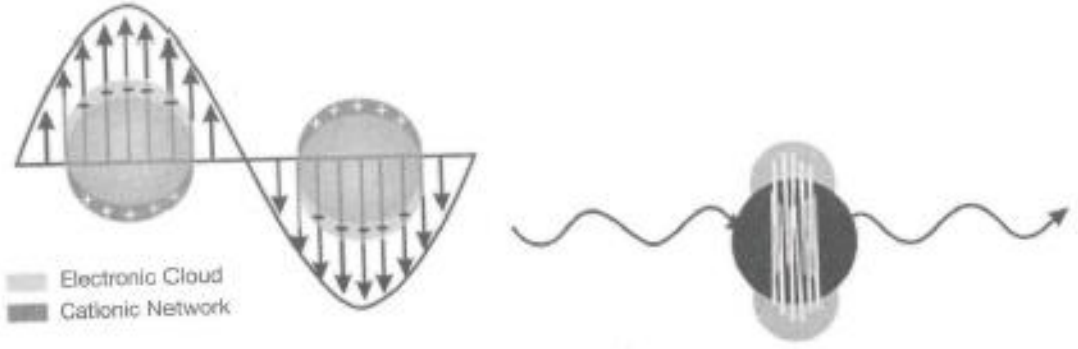
Material properties (such as optical, Absorption, shown here) change with the size of the gold nanoparticle.

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Properties of nano materials

Optical property: Surface Plasmonic Resonance (SPR)

- As the size of the particle decreases surface area increases and electron density on surface increases.
- The surface electrons oscillate with certain frequency. It is called plasmons.
- When light falls on the surface and if the frequency of incident light matches with the frequency of plasmons, then it is under resonance.
- Hence it is called Surface Plasmonic Resonance.



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Classification of nano materials

Nano scale materials are divided into four category,

1.Zero dimensional (0D): Nanomaterials are less than 100 nm in all three dimensions.

2.One dimensional (1D): Nanomaterials are less than 100 nm in any two dimensions.

3.Two dimensional (2D): Nanomaterials are less than 100 nm in anyone dimensions.

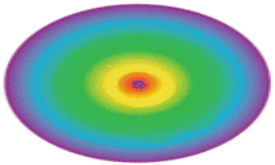
4.Three dimensional (3D): Nanomaterials are aggregations of any of the above (0, 1, 2D) dimensional nanomaterial with greater than 100nm in all the three

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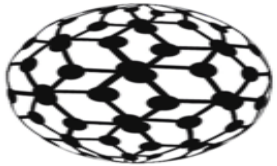
Classification of nano materials

NMs classification based on dimensionality

0D
Nanospheres, clusters



Quantum dots



Fullerenes

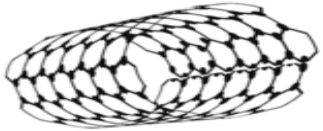


Gold nanoparticles

1D
Nanotubes, wires, rods



Metal nanorods, Ceramic crystals



Carbon nanotubes, Metallic nanotubes

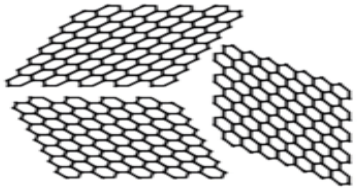


Gold nanowires, Polymeric nanofibers, Self assembled structures

2D
Thin films, plates, layered structures



Carbon coated nanoplates



Graphene sheets



Layered nanomaterials

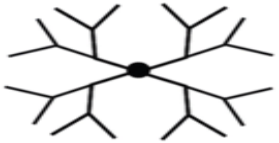
3D
Bulk NMs, polycrystals



Liposome



Polycrystalline



Dendrimer

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Synthesis of nano materials

General methods of synthesis of nanomaterials	
CHEMICAL METHODS (Bottom Up)	**PHYSICAL METHODS** (Top down)
Combustion Synthesis	Inert gas condensation
Hydrothermal	Sputtering
Sol-gel method	Molecular beam epitaxy
Micelles micro emulsion	Lithography
Single crystal growth	Ion beam technique
Colloidal methods	
Chemical vapor deposition	

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Solution Combustion Method

- **Combustion method:** is a low temperature, time saving, energy efficient, self propagating method, involving **spontaneous exothermic redox reaction** between **metal nitrate and organic fuel**, used to prepare nano-materials.

- **Principle:** The method is based on the utilization of heat energy produced during the exothermic spontaneous redox reaction between an **oxidizer (metal nitrate)** and a **reducing agent (organic fuel)**. The oxidizer can be of any metal nitrates and reducing agents may be organic fuels, such as glycine, oxalic acid, urea, hexamine, sugar, EDTA, Dextrose etc.

Metal nitrate + Fuel ⟶ Nano metal oxide + Gasses

Solution Combustion Method

- The Fuel required is calculated by using the following formula

$$\frac{\text{Wt. of fuel}}{\text{Mol. Wt. of fuel}} \times \text{Valency of fuel} = \frac{\text{Wt. of metal nitrate}}{\text{Mol. Wt. of metal nitrate}} \times \text{Valency of metal nitrate}$$

For the calculation of valency of fuel and metal nitrate, the valency to be considered for the elements such as of N, O, H, C, and M^{n+} are 0, -2, +1, +4 and +n respectively.

Procedure:

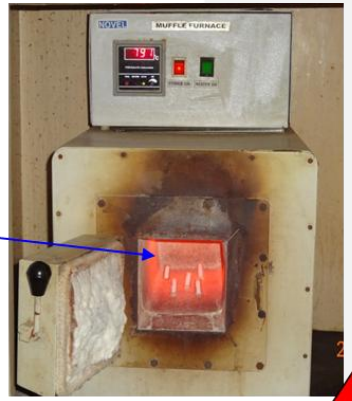
Step1: Weighing and mixing of the reactants:



Solution Combustion Method

Step 2: Evaporation and combustion:

The uniformly mixed solution is kept in a furnace maintained at 500 °C





Evaporation (boiling) of water

Spontaneous/vigorous combustion occurs and propagates throughout the redox mixture





Final nano powder of corresponding metal oxide

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Advantages of Solution Combustion Method

- It is an easy and fast process that uses relatively simple equipment.
- High-purity products can be easily prepared by using this method.
- Composition, structure, homogeneity, size and stoichiometry of the products can be controlled.
- High exothermicity of the metal nitrate–fuel reaction permits incorporation of desired quantity of dopants in the various hosts of industrially useful phosphors.
- This is the cheap method as compared to conventional solid state method.

Low temperature solution combustion (SCS) method

Example:

Synthesis of MgO by combustion method:

Chemicals: Magnesium nitrate ($Mg(NO_3)_2.6H_2O$, (Mol. Wt=256.41) any organic fuel (say urea; NH_2CONH_2; Mol. Wt = 60) and distilled water.

Calculation of fuel:

The valency of urea, N_2H_4CO= 2(0)+4(+1)+1(+4)+1(-2)= 6

The valency of $Mg(NO_3)_2.6H_2O$= 1(+2)+2(0)+ 6(-2)+6(0)= 10

The amount of urea for 5.0 g of $Mg(NO_3)_2$ is calculated as follows.

$$\frac{\text{Wt. of fuel}}{\text{Mol. Wt. of fuel}} \times \text{Valency of fuel} = \frac{\text{Wt. of metal nitrate}}{\text{Mol. Wt. of metal nitrate}} \times \text{Valency of metal nitrate}$$

Wt of urea = (5x10x60)/(256.41x6)= 1.95g

Step-I: 5g of magnesium nitrate and 1.95g of urea are dissolved in minimum amount of water .

Step-II: The solution kept in pre heated furnace at 500C.

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Commonly used Chemicals

Choose the chemical such that from which we can get the resultant component.

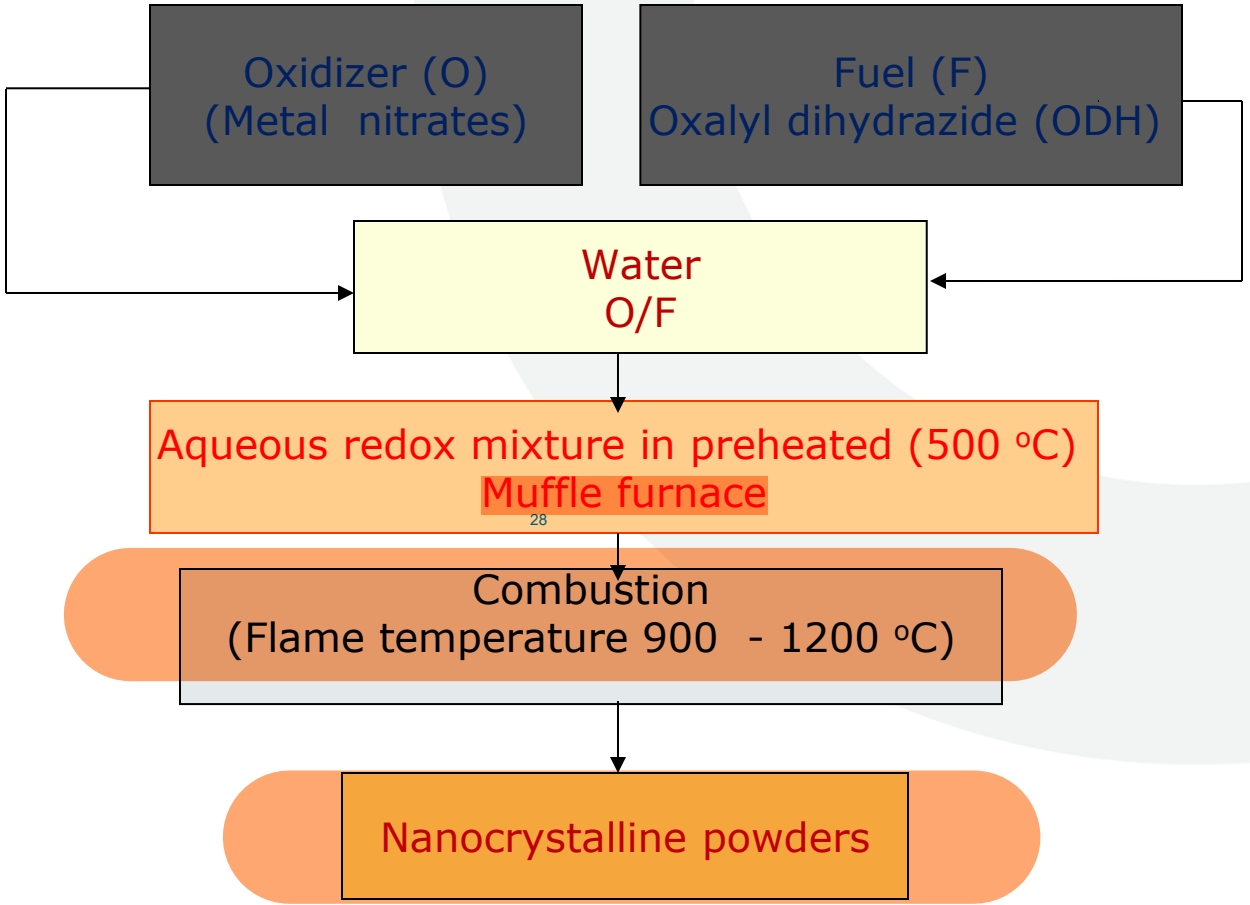
- Ex: Oxidizer
  - For CuO → $Cu(NO_3)_2$ Copper N
  - For ZnO → $Zn(NO_3)_{2-}$ Zinc Nitr

- Choose the fuel such as containing carbon and hydrogen main components
- Ex:
  $C_2H_5NO_2$ → Glycine
  $C_6H_8O_6$ → Ascorbic acid



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The flow chart for Combustion Method



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Solution combustion (SCS) method

Examples of nano metal oxides



1 atom % Co^{2+} in $SrAl_2O_4$



1 atom % Cr^{3+} in α- Al_2O_3

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Sol-Gel Method

- Sol-gel process involves the conversion of precursor solution (usually metal salts or metal alkoxide) into a nano-structured inorganic solid through inorganic polymerization reactions catalyzed by water.

Reactions involved:

Hydrolysis

$M\text{-}OR + H_2O \longrightarrow M\text{-}OH + ROH$

Metal alkoxide — Metal hydroxide

Condensation

$M\text{-}OR + H\ O\text{-}M \longrightarrow M\text{-}O\text{-}M + R\text{-}OH$

Steps involved:

Step 1: Preparation of homogeneous solution either by dissolution of metal salt in water or metal alkoxide in an organic solvent (usually alcohol) that is miscible in water.

Step 2: 'Sol' formation by hydrolysis

It involves the conversion of homogeneous solution into a "sol". During hydrolysis, the alkoxide groups (-OR) are replaced via the nucleophilic attack of the oxygen atom of a water molecules, results in release of alcohol and formation of metal hydroxide (sol). A stable dispersion of colloidal particles of precursors in a solvent is known as sol)

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Sol-Gel Method

Step 3: 'Gel' formation by condensation

The colloidal solution is kept for aging. During aging condensation reaction between two metal hydroxyl/alkoxy species leads to M-O-M bonds with the release of H_2O/R-OH. This condensation process continues and finally results in a "gel", an interconnected, a rigid and porous inorganic network covered completely with liquid phase. This transformation is called **Sol-Gel transition**.

Step 4: Drying of gel:

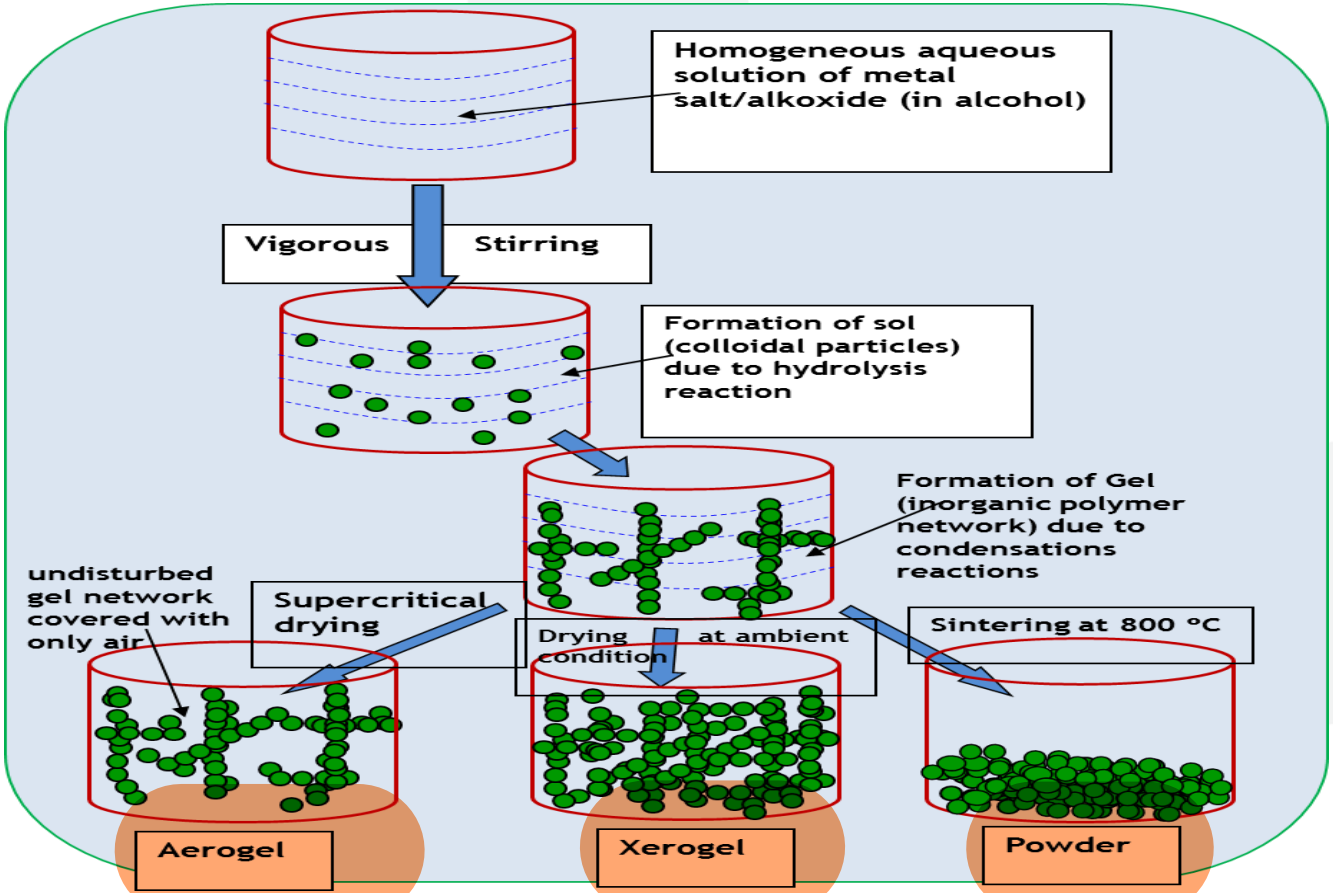
It involves removal of liquid phase from gel network. There are different ways of drying gel:

a) If the gel medium is dried, by removing the liquid solvent (under hyper critical conditions) without destroying the gel network, **"aerogel" is** produced.

b) If the solvent is dehydrated under ambient conditions (removal of R-OH groups), **"xerogel"** are produced.

c) If the gel network is sintered at high temperature (800ºC), densification, decomposition of gel results in complete collapse of gel net into powder.

Sol-Gel Method



Sol-Gel Method

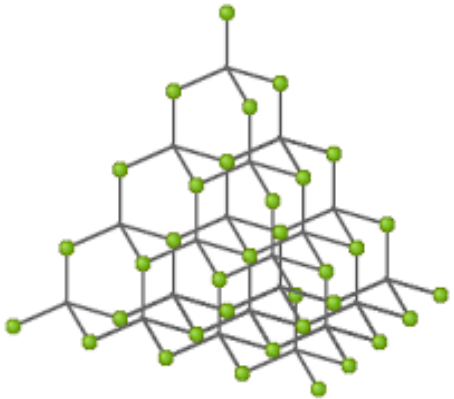
Advantages of the technique.

- It is a simple and economic technique, as the fabrication does not need expensive machinery or apparatuses.
- It offers a high flexibility as one can produce materials with a wide range of stoichiometry and additional dopants.
- It allows the fabrication of high-quality coatings.
- The starting materials are easy to obtain, not expensive and available in a high purity.
- Nanomaterials of variety of shapes can be prepared.

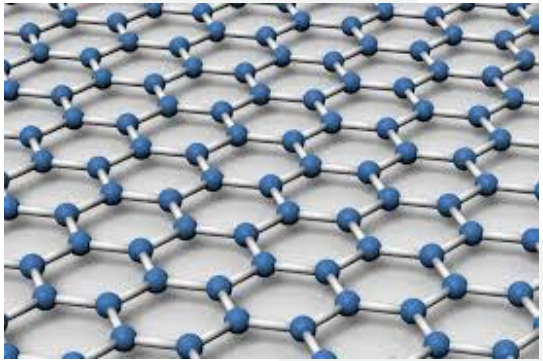
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Carbon Nano materials

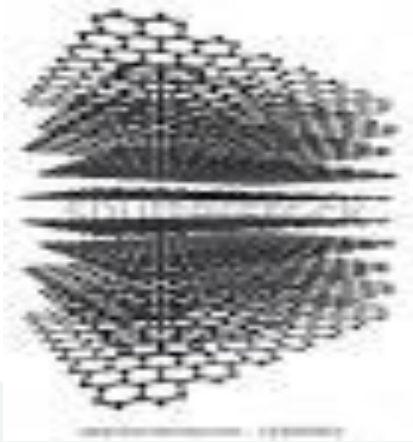
➢ Allotropes of carbon



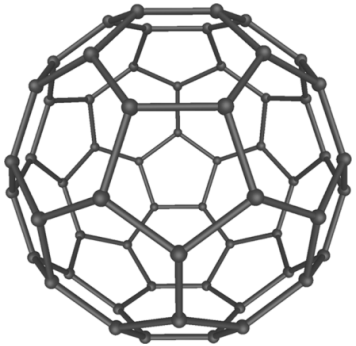
Diamond



Graphene



Graphite

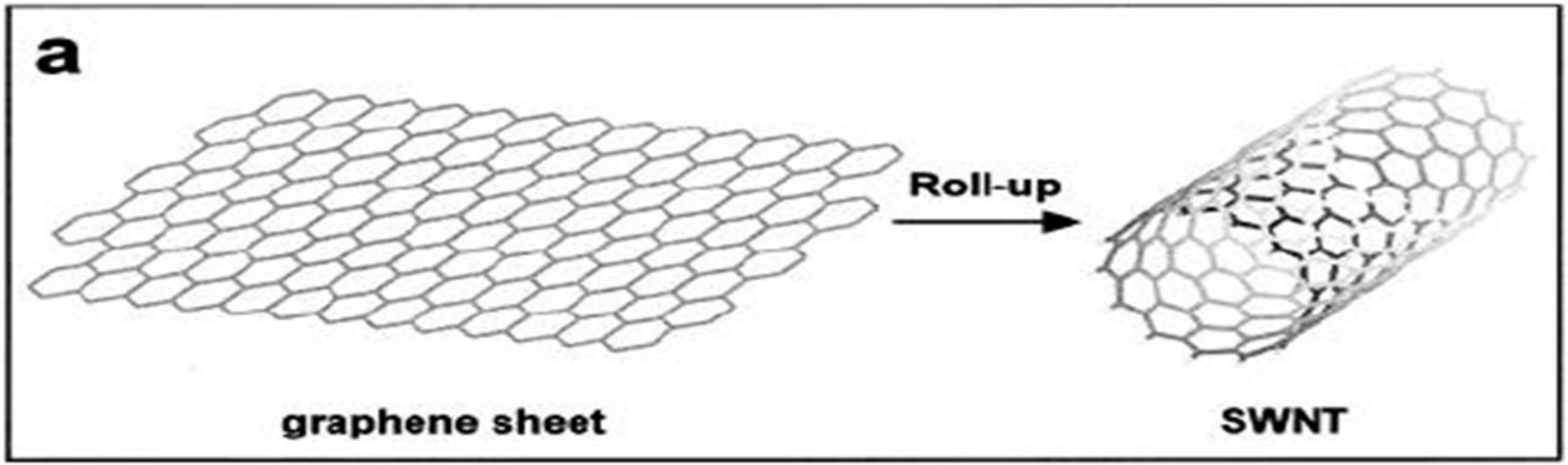


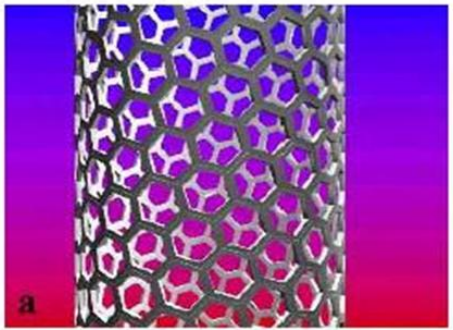
Fullerene

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Carbon Nanotubes

- CNT: Rolling-up a graphene sheet to form a tube



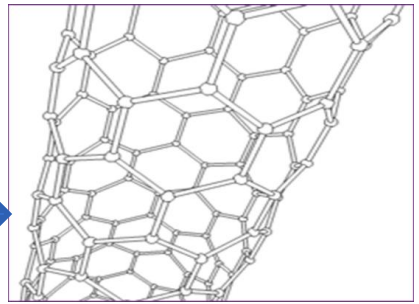


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Carbon nano tubes

Unique properties of CNT's

Carbon nanotubes, composed of interlocking carbon atoms, are 1000x thinner than an average human hair – but can be 200x stronger than steel.



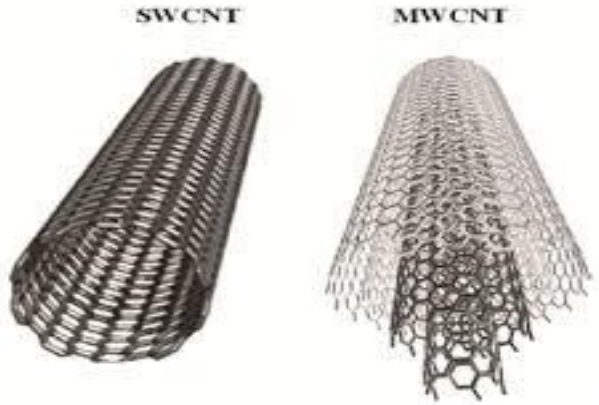
CARBON NANO-TUBES

- Extra ordinary strength
- Thermal Properties
- Good Electrical Properties

Carbon nanotubes

CARBON NANOTUBES

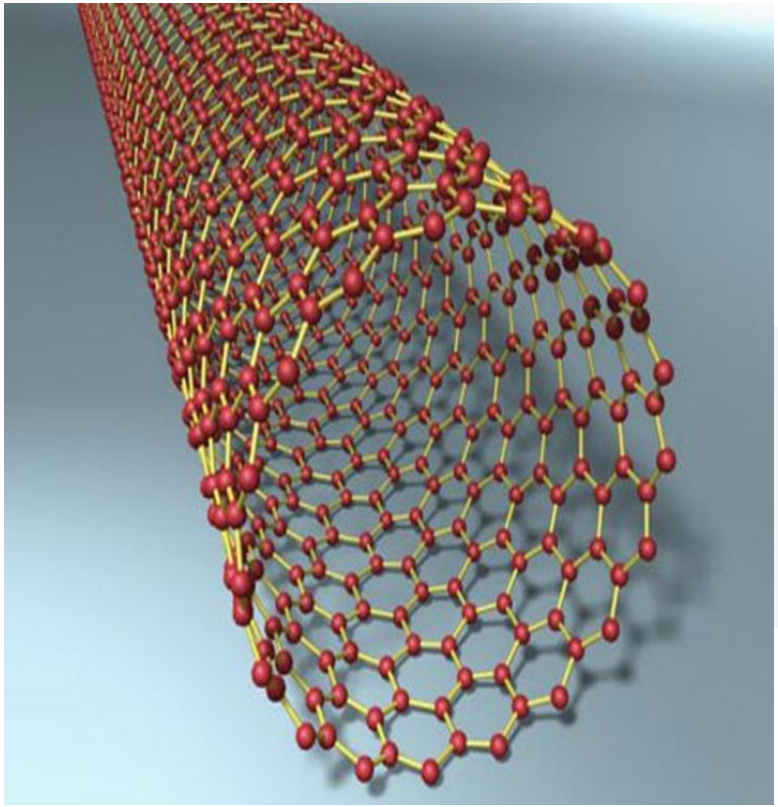
- SINGLE-WALLED(SWNT)
- MULTI-WALLED(MWNT)



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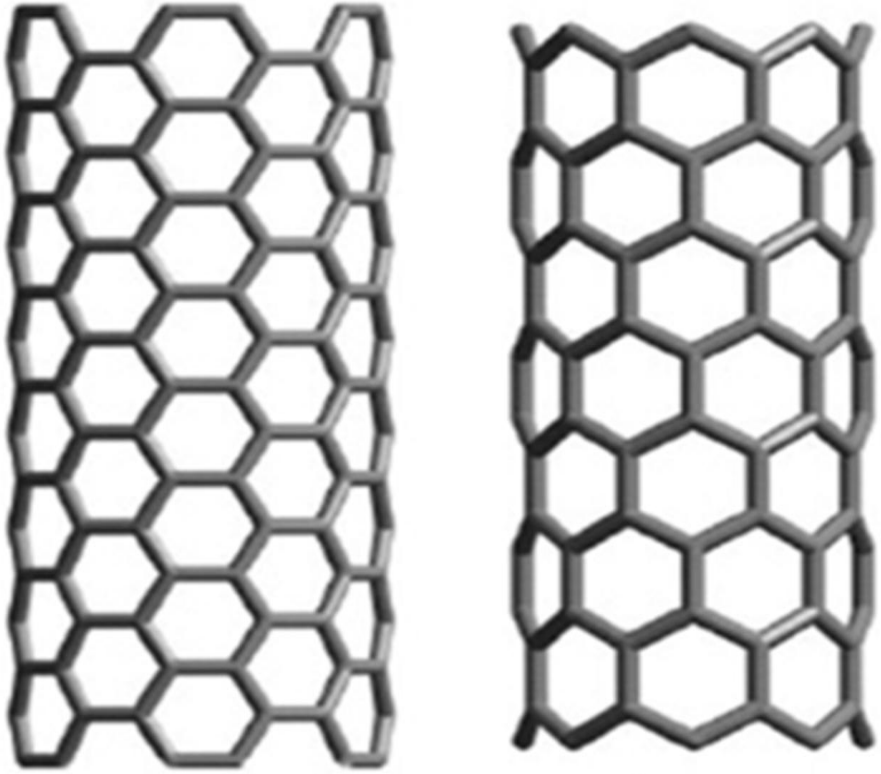
Single walled Carbon nanotubes

- Diameter of close to 1 nanometer
- Tube length that can be many millions of time longer



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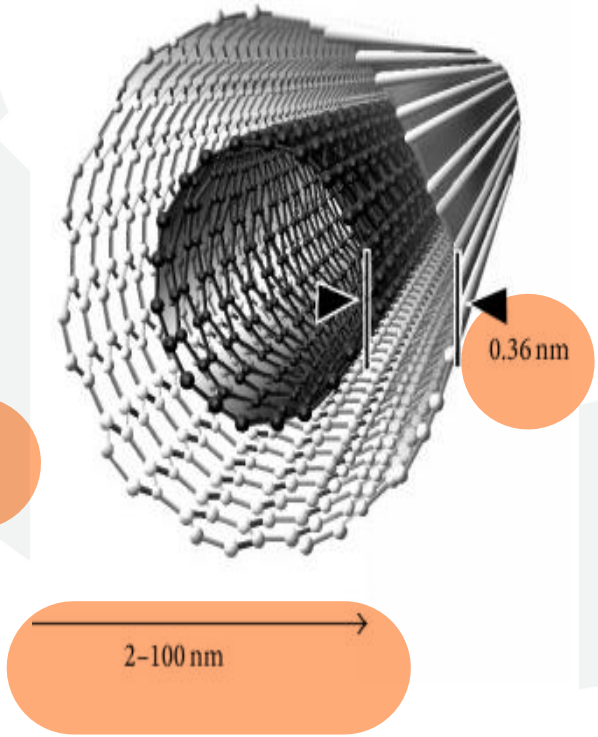
Types of Single walled Carbon Nano-tubes: (SWCNT)



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Types of Multi walled Carbon Nano-tubes: (MWCNT)

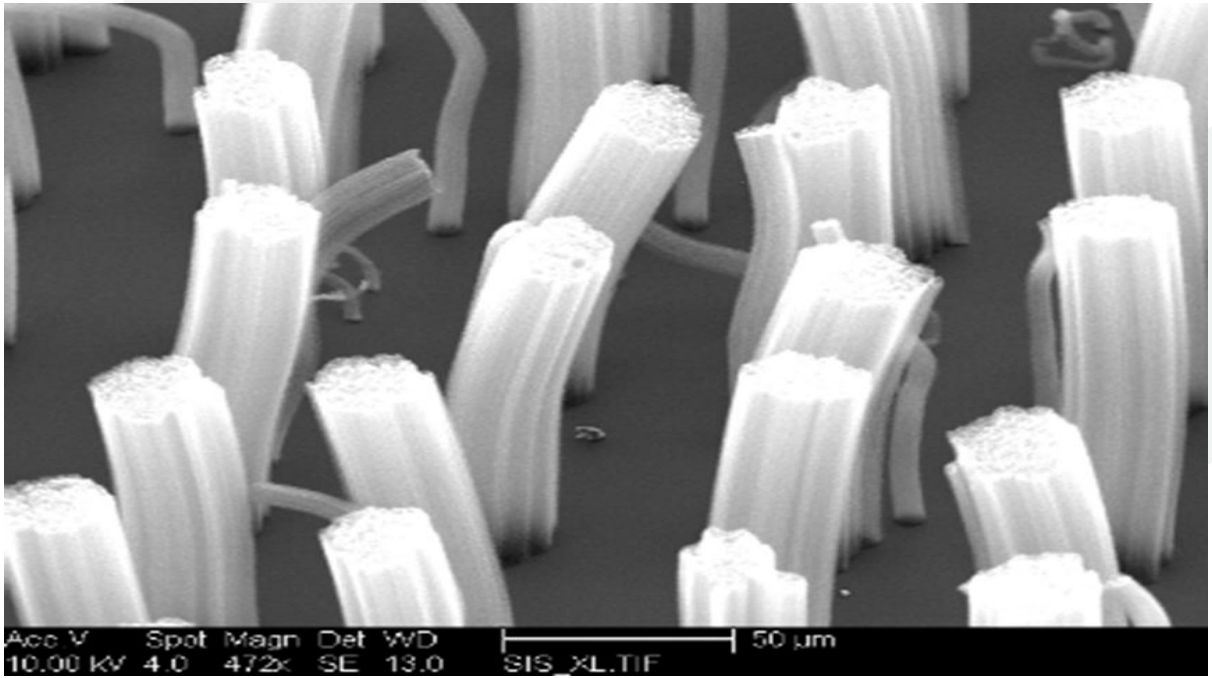




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Methods of preparation of CNT's

- Chemical Vapor Deposition (CVD)
- Arc Discharge
- Laser Ablation
- Ball Milling



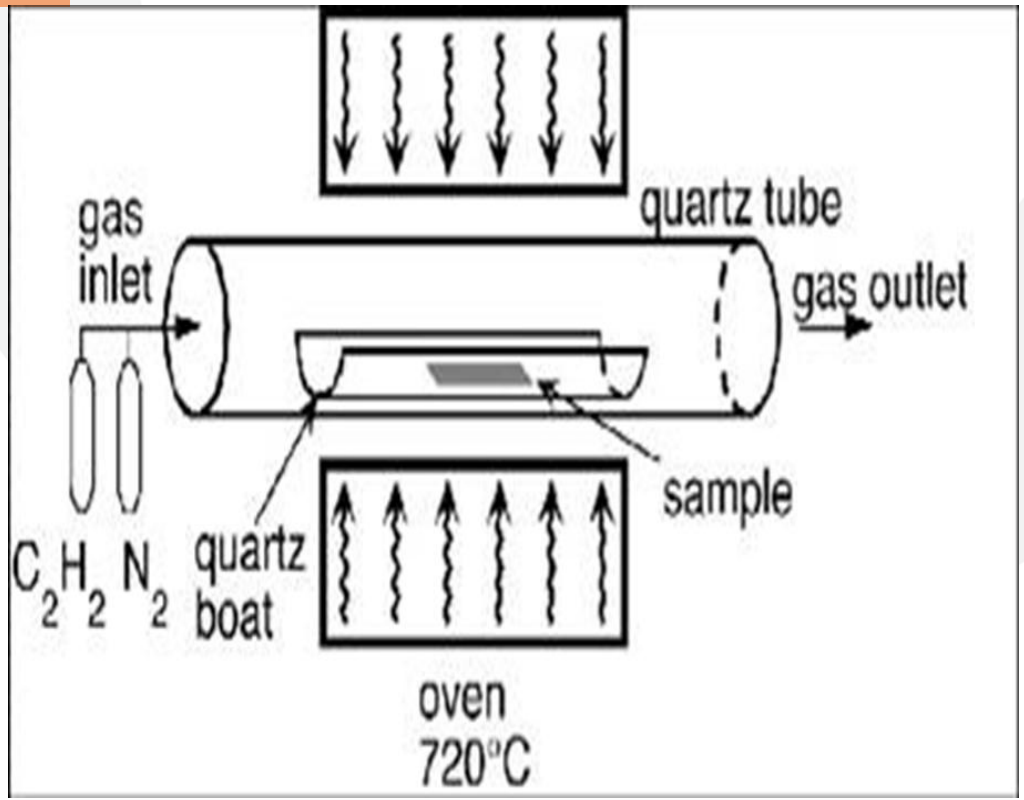
Chemical Vapor Deposition Method (CVD)

- One of the most common methods of carbon nanotube synthesis.
- Temperature between 650 – 1200ºC.
- Yield is usually about 30%.

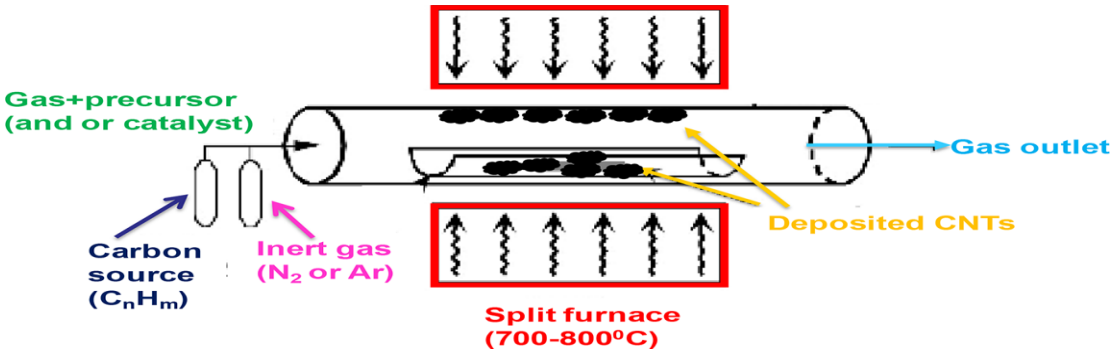
Source of carbon: Methane, Carbon monoxide, Acetylene gasses.

Usually, a silicon plate coated with iron particles is the substrate.

After energy transfer, the carbon molecule binds to the substrate.



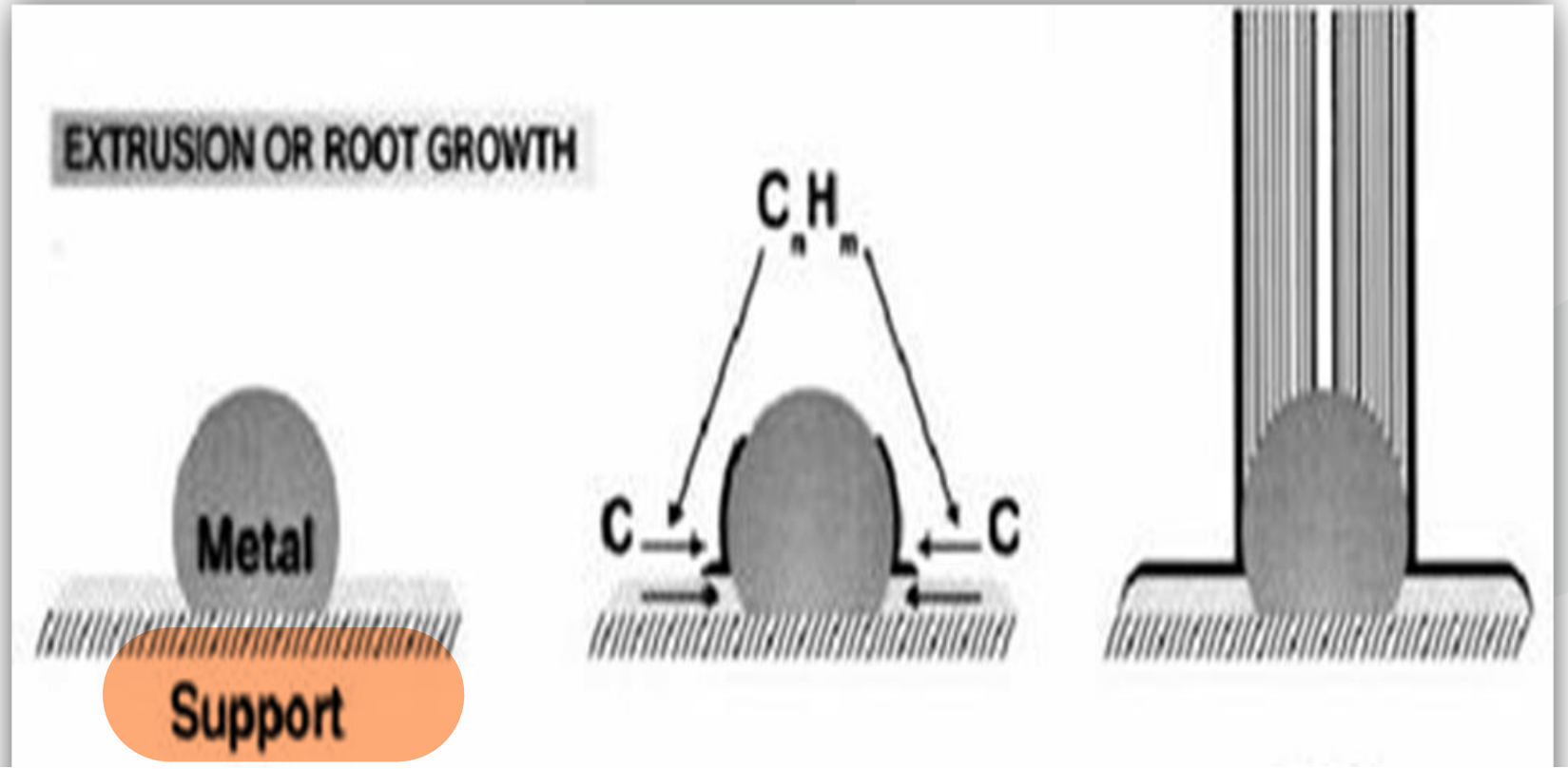
- Precursors- Hydrocarbons
- Substrate- carbon, quartz covered with metal catalyst like Co, Ni, Sr
- Reactor- It consists of a quartz tube of 30mm diameter and 1000mm length
- Step1- Hydrocarbon along with argon gas(carrier gas) is fed into reaction chamber maintained at high temperature 600-1200 C and under vacuum
- Step 2- At high temperature, hydrocarbons undergo decomposition C-H bonds break and carbon atoms are formed. They bind together to form carbon nanotube (due to Vander Waals forces of attraction)
- Step 3- These carbon nanotubes grow on the catalyst and are collected after cooling.
- During this process , volatile byproducts are also produced which are removed by
- gas flow through reaction chamber.
- H2 is used to maintain reducing atmosphere.



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Chemical Vapor Deposition Method (CVD)

- Growing of nano-tube on the surface of a substrate:



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Properties of CNT's

- Carbon-carbon bonds are one of the strongest bond in nature, **Composed entirely of sp^2 bonds.**
- 132,000,000:1 Length-To-Diameter Ratio.
- Diameter of 3 to 9 nm.
- Lengths in the millimeter range.
- Extremely high Young's modulus.

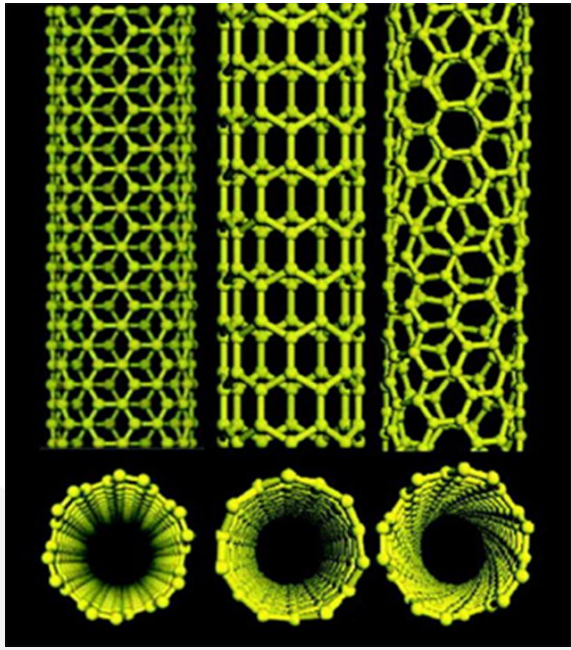
Material	Young's modulus (GPa)
Steel	190-210
SWNT	1,000+
Diamond	1,050-1,200

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Properties of CNT's

Electrical Properties

- If the nanotube structure is armchair, then the electrical properties are metallic.
- If the nanotube structure is chiral then the electrical properties can be semiconducting with a very small band gap, otherwise the nanotube is a moderate semiconductor.
- In theory, metallic nanotubes can carry an electrical current density of 4×10^9 A/cm^2 which is more than 1,000 times greater than metals such as copper.



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Applications of CNT's

- **Coating and Films:**
- CNT-based transparent conducting films.
- Alternative to expensive indium tin oxide (ITO).
- Flexible, and not brittle.
- Application in.
  - Displays
  - Touch screen devices
  - photovolatics



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Applications of CNT's

Nanotubes In Efficient Solar Cells

- Scientists have developed the 'blackest black' colour using carbon nanotubes
- Use of carbon nanotubes in solar cells could vastly improve their efficiency.



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Applications of CNT's

Nanotubes In Sporting Equipment

- Badminton racquet manufacturer Yonex incorporates carbon nanotubes into their cup stack carbon nanotubes racquets.
- American baseball bat manufacturer Easton Sports has formed an alliance with a nanotechnology company Zyvex to develop baseball bats incorporating carbon nanotubes.
- Tennis racquets also incorporate carbon nanotubes.



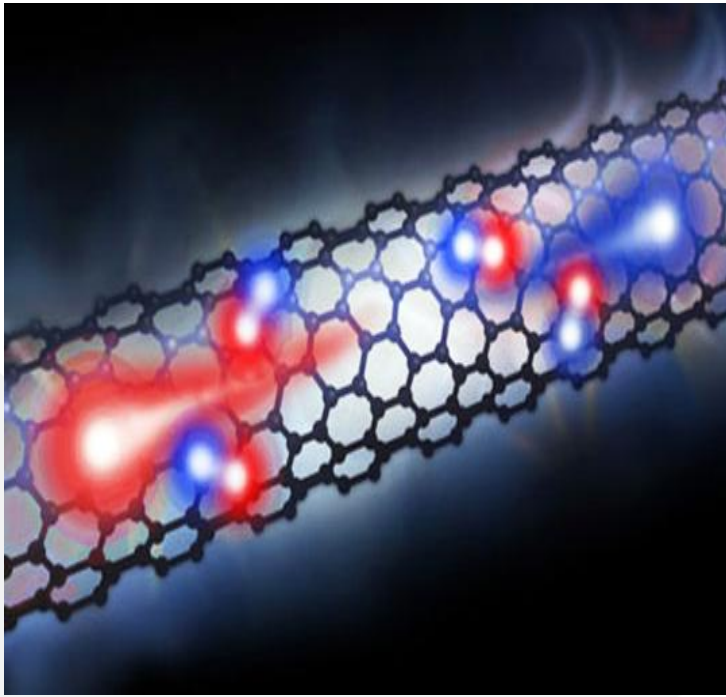
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Applications of CNT's

In cancer treatment

- CNTs are tubular materials with nanometer-sized diameters and axial symmetry, giving them unique properties that can be exploited in the diagnosis and treatment of cancer.

- CNTs have the potential to deliver drugs directly to targeted cells and tissues. Alongside the rapid advances in the development of nanotechnology-based materials, elucidating the toxicity of nanoparticles is also imperative



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Carbon nano tubes (CNT's)

Hazards:

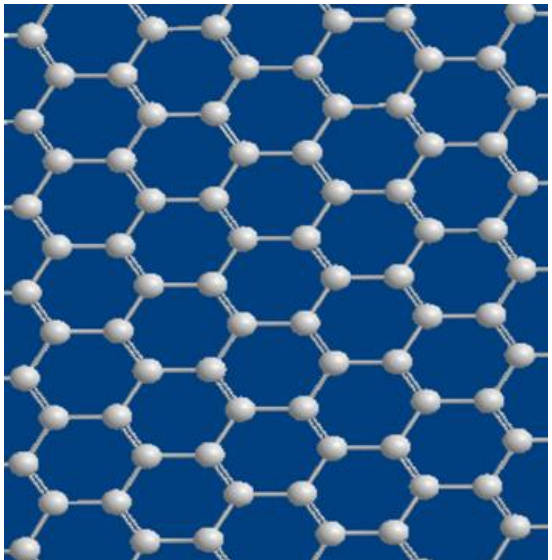
- According to scientists at the National Institute of Standards and Technology, carbon nanotubes shorter than about 200 nanometers readily enter human lung cells like the way asbestos does and may pose an increased risk to health.

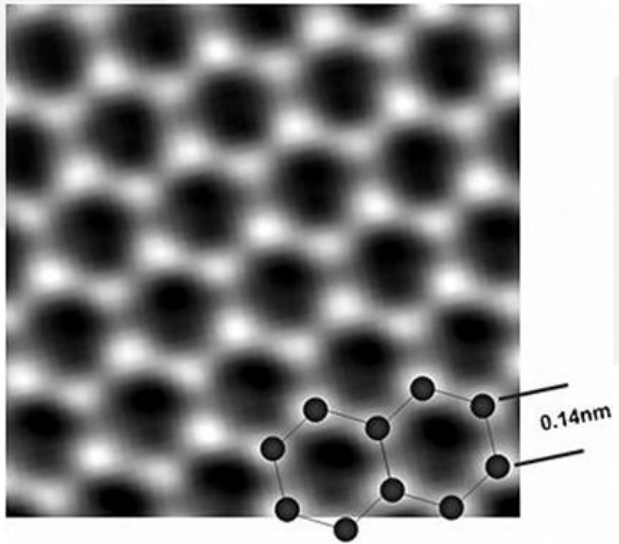
- Carbon nanotubes along with most of the nanotechnology, are an unexplored matter, and many of the possible health hazards are still unknown.

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GRAPHENE

- A single sheet of graphite is called graphene.
- It is sp^2 hybridized → covalently bonded.
- The bonding between the graphene sheets in graphite is of van der Walls type.
- 2-dimensional, crystalline allotrope of carbon

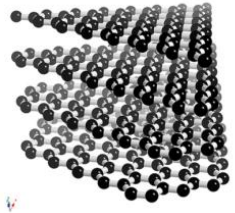




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Synthesis of graphene

Modified Hummer's method



Graphite

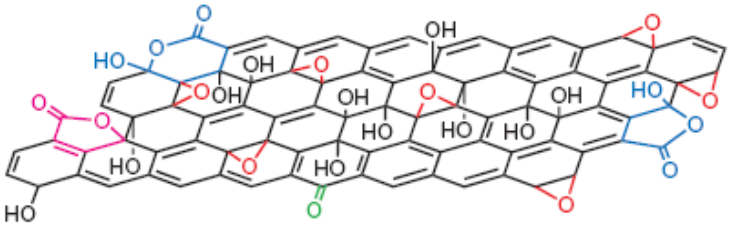
Oxidation

H_2SO_4/ $KMnO_4$
H_2SO_4/$KClO_3$
Or
$H2SO_4$/HNO_3

Graphite Oxide

H_2O

Ultrasonication (exfoliation)



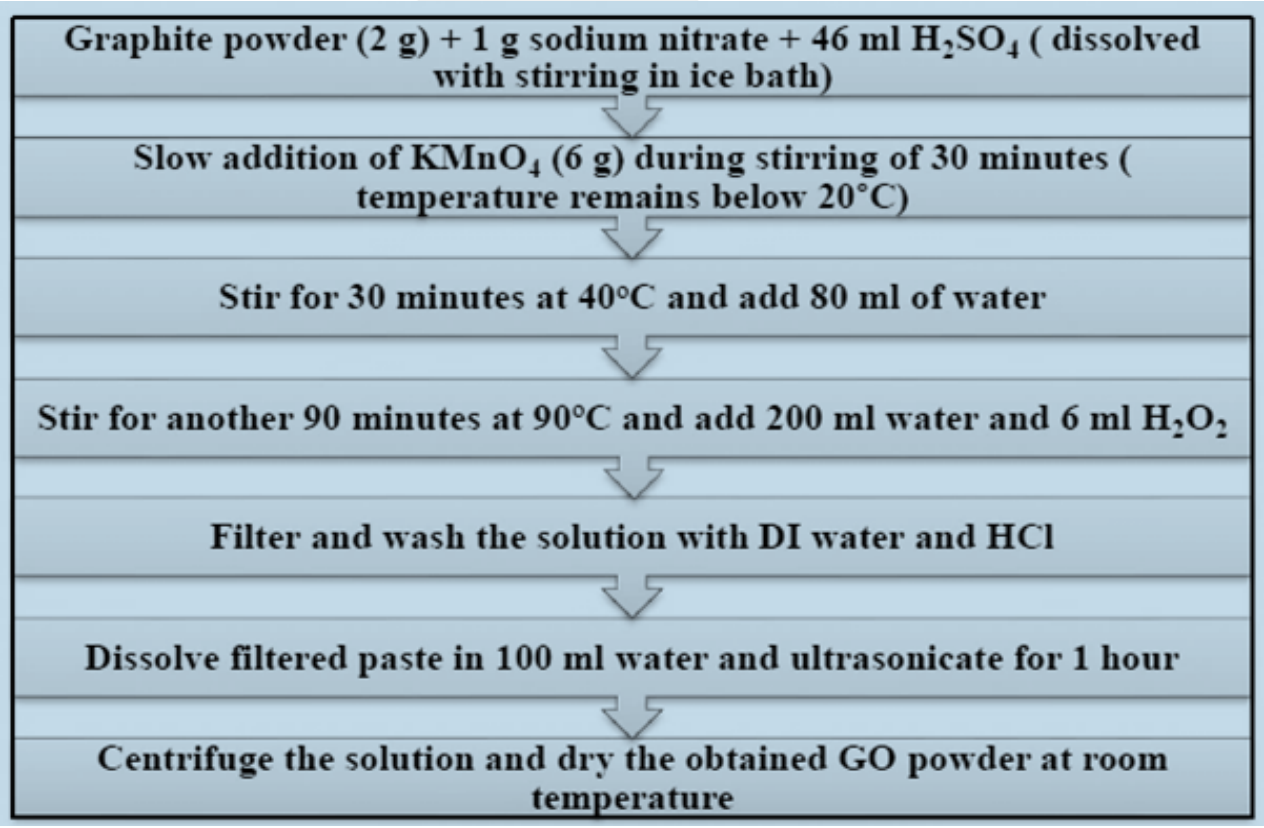
Graphene Oxide
monolayer or few layers

Reducing agetnt

Reducing Graphene Oxide (RGO)

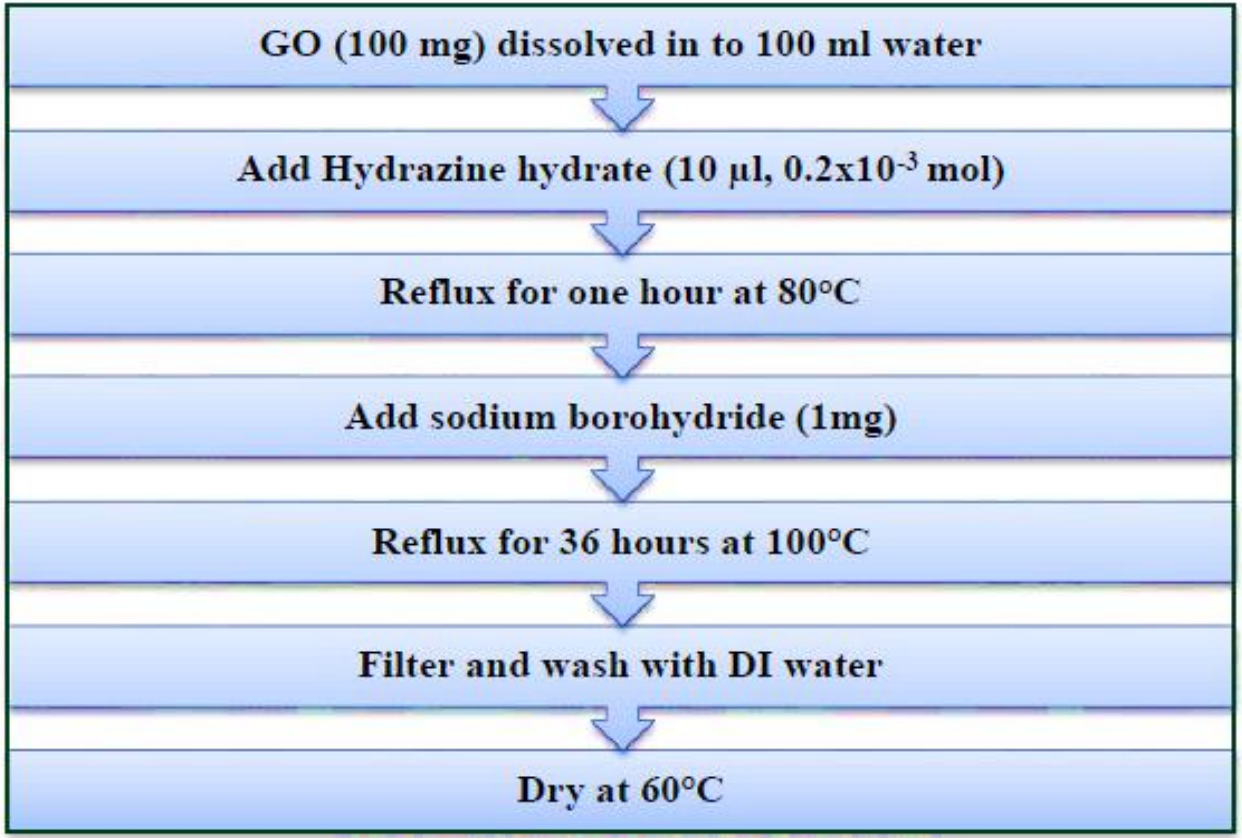
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Stage I - Synthesis of GO (graphene oxide) from graphite powder

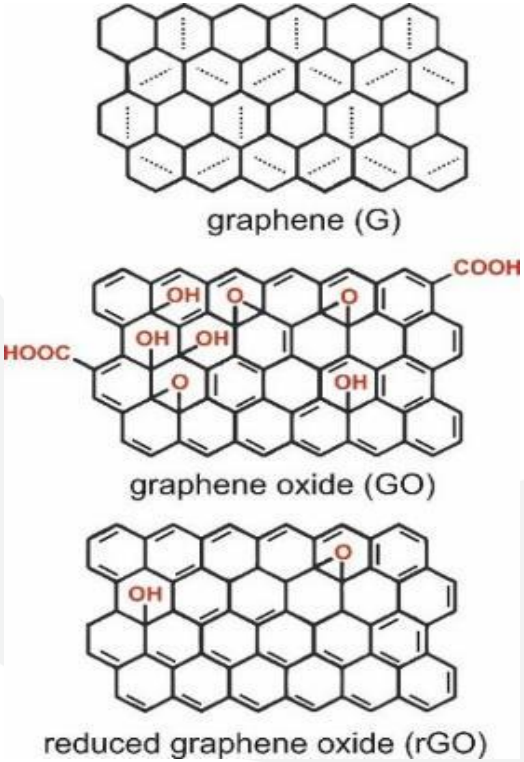


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Stage-II: Synthesis of Graphene from GO (graphene oxide)




Graphite
9 KMnO4
H2SO4
Graphite oxide
H2O2
PRODUCTION LINE
Centrifugation
H2O
HCl
rGO
GO
NaBH4
95°C
Stirring
Centrifugation
H2O
Sonication
0.5 hours
Driying
80°C
24 hours


graphene (G)
COOH
HOOC
OH
O
graphene oxide (GO)
reduced graphene oxide (rGO)

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Properties of Graphene

- Conductive: Electrons are the particles that make up electricity. So when graphene allows electrons to move quickly, it is allowing electricity to move quickly. It is known to move electrons 200 times faster than silicon because they travel with such little interruption. It is also an excellent heat conductor. Graphene is conductive independent of temperature and works normally at room temperature.
- Strong: As mentioned earlier, it would take an elephant with excellent balance to break through a sheet of graphene. It is very strong due to its unbroken pattern and the strong bonds between the carbon atoms. Even when patches of graphene are stitched together, it remains the strongest material out there.
- Flexible: Those strong bonds between graphene's carbon atoms are also very flexible. They can be twisted, pulled and curved to a certain extent without breaking, which means graphene is bendable and stretchable.
- Transparent: Graphene absorbs 2.3 percent of the visible light that hits it, which means you can see through it without having to deal with any glare.

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Applications of graphene

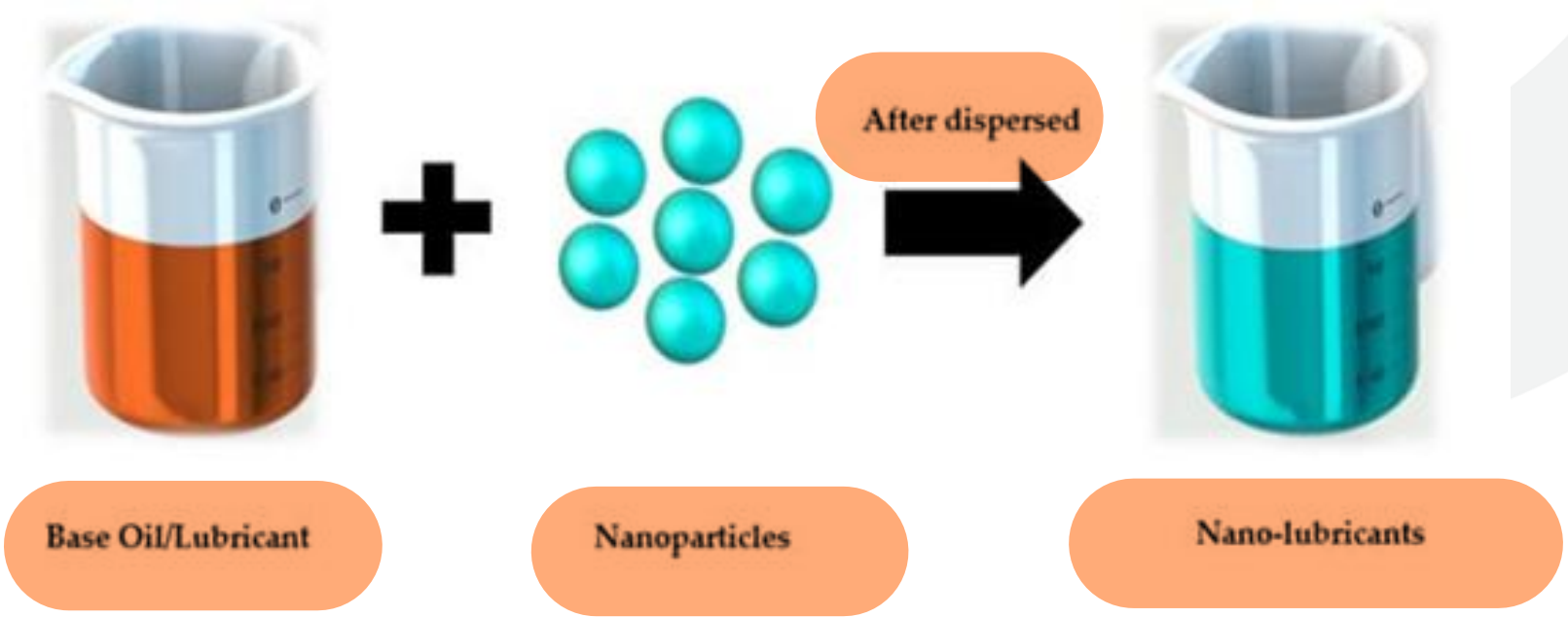
- OLED Techonologies
- Body Armour
- Lightweight Aircraft/vehicles
- Photovoltaics
- Superconductor/battery
- Filtration

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Nano Lubricants

- Lubricants are substances used to reduce friction and wear between moving parts, ultimately improving the efficiency and lifespan of mechanical systems.

- Nano lubricants are the dispersion of nanoparticles in a base oil.



Nano Lubricants

- Nanoparticles used in nano lubricants will provide several advantages due to their unique properties at such a small size.
- **Improved Lubrication Performance:** Nano lubricants can reduce friction more effectively than traditional lubricants. The small size of nanoparticles allows them to access and fill microscopic irregularities on the surfaces of moving parts, leading to smoother interactions.
- **Enhanced Heat Transfer:** The high surface area-to-volume ratio of nanoparticles can improve heat transfer within the lubricant, aiding in dissipating heat from the contact zones.
- **Reduced Wear and Tear:** Nano lubricants can form protective layers on surfaces, preventing direct metal-to-metal contact and reducing wear and tear. This is especially beneficial in high-stress conditions.
- **Increased Load-Carrying Capacity:** The addition of nanoparticles can enhance the lubricant's load-carrying capacity, making it suitable for applications involving heavy loads.

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Nano Lubricants

- **Improved Fuel Efficiency:** By reducing friction and wear, nano lubricants can contribute to improved energy efficiency in various applications, such as internal combustion engines.

- **Extended Equipment Lifespan:** The protective qualities of nano lubricants can help extend the lifespan of mechanical components by minimizing wear-related damage.

- **Customizable Properties:** Nano lubricants can be engineered to exhibit specific properties, such as viscosity, thermal conductivity, and stability, by adjusting the type and concentration of nanoparticles.

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Nano Lubricants

However, there are challenges associated with the development and application of nano lubricants:,

- **Nanoparticle Dispersion:** Achieving uniform dispersion of nanoparticles in the base lubricant can be challenging, as nanoparticles tend to agglomerate. This can lead to uneven lubrication and reduced effectiveness.

- **Stability:** Nanoparticles can settle over time, causing the properties of the lubricant to change. Maintaining stability is crucial for consistent lubrication performance.

- **Compatibility:** Nano lubricants may not be compatible with all materials and components, potentially leading to unintended interactions or damage.

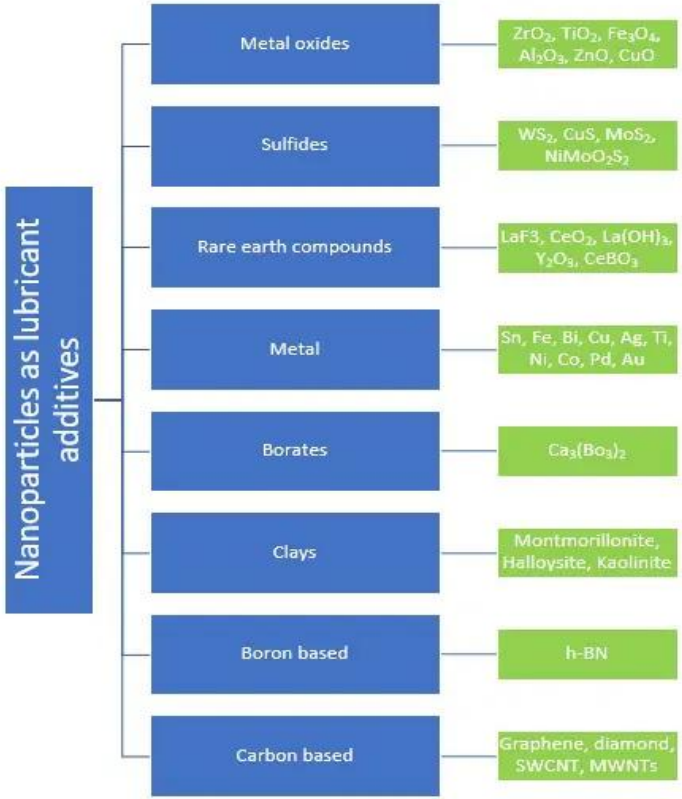
- **Health and Environmental Concerns:** The potential health and environmental impacts of nanoparticles, particularly when they are released into the environment, are subjects of research and regulation.
- **Cost:** The production and integration of nanoparticles into lubricants can be costly, potentially limiting their widespread adoption.

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Nano Lubricants

There are three basic types of nano lubricant additives:

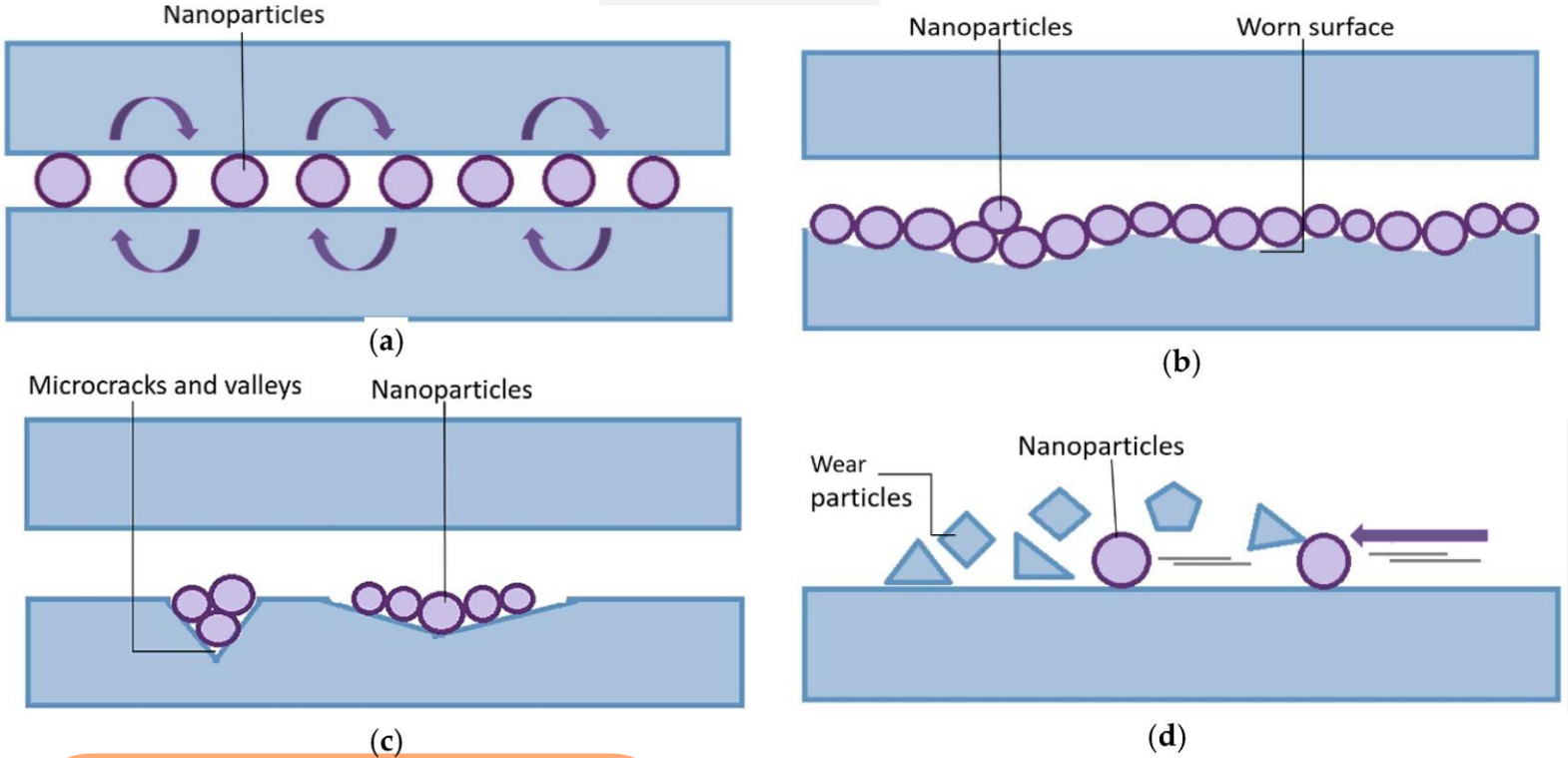
1. Nano metal-based additives
2. Nano carbon-based additives
3. Nano composite-based additives



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Nano Lubricants

Lubrication mechanisms of nano lubricant additives

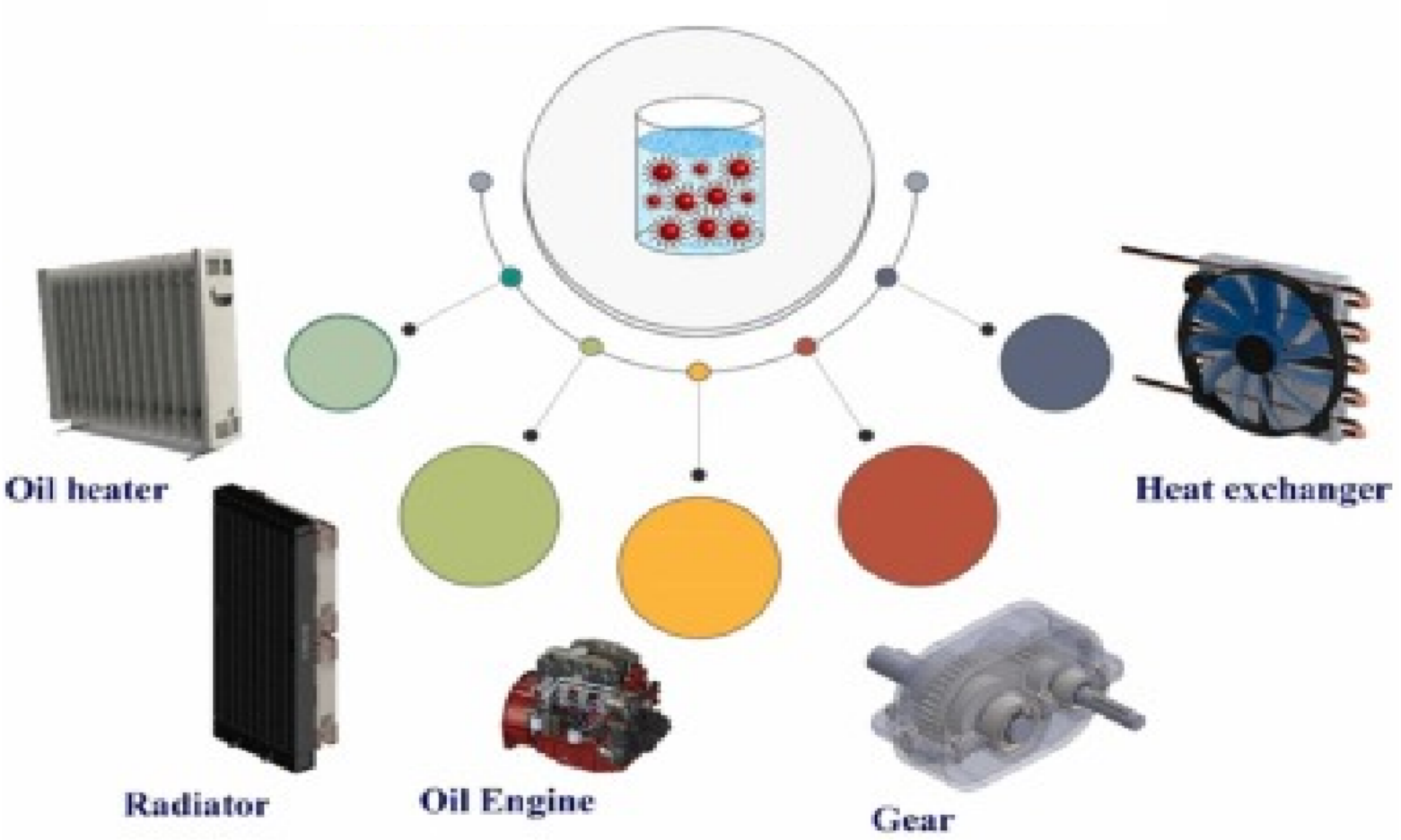


a) Rolling effect
c) Mending effect

b) Protective film
d) Polishing effect

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Applications of Nano Lubricants



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